Table of Contents

Generate a chirp signal with frequency increasing 1
from f0 to (f0 + fstep time) over time seconds 1
Add code here to define the chirp signal y = cos(angle(t)) 1

Generate a chirp signal with frequency increasing

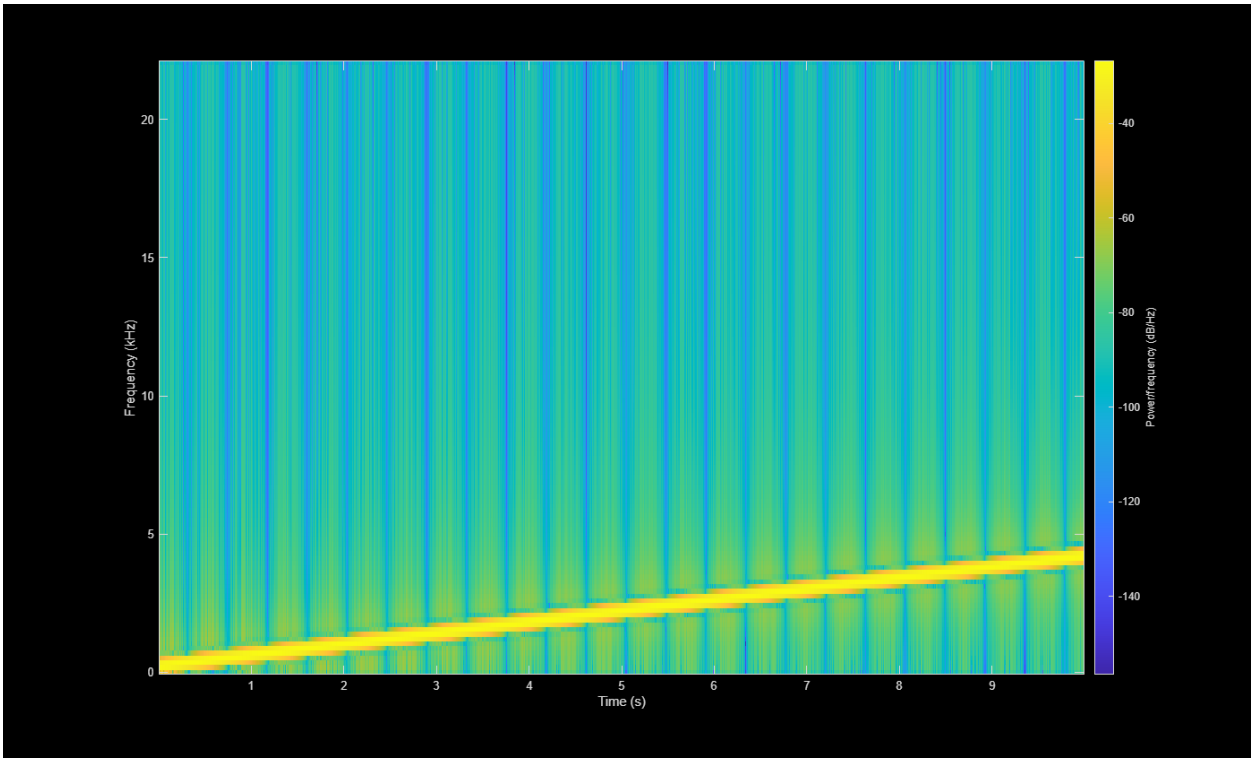
from f0 to (f0 + fstep time) over time seconds

```
time = 10;
f0 = 220;
fstep = 400;
fs = 44100;
Ts = 1 / fs;
t = 0 : Ts : time;
```

Add code here to define the chirp signal y = cos(angle(t))

```
angle = 2*pi*f0*t+pi*fstep*t.^2;
y = cos(angle);
sound(y, fs);
% I hear one continuous signal with the pitch/frequency increasing as time
% increases. It sounds like its getting louder too because the frequencies
% are sharper and pierce more.

figure;
spectrogram(y, hamming(256), 128, 256, fs, 'yaxis');
% we have a continuous audio signal in which the base frequency increases
% proportionally to time. The actual frequency is depicted by the yellow
% and the orangish areas are caused by the overlap between sampling
% segments. Within the 10 seconds, it goes from 220Hz to 4186Hz
```



Published with MATLAB® R2025b